RAHUL A

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Summary

Software Engineer building Python backend systems, including REST and real-time APIs, authentication, Redis-based concurrency control, RAG and LLM applications, and logistics automation, with prior experience in Java ERP and billing systems.

Experience

Software Developer (Python) Mar 2025 – Present
Zoondia Technologies *Trivandrum, India*

- Built CargoXFlow, an import/export logistics platform, owning the backend from schema design to real-time features and locking.
- Built AI-powered recruitment automation and RAG-based chatbot systems covering resume parsing, email processing, candidate matching, and document-grounded Q&A.
- Designed and documented authentication and business APIs in OpenAPI (Swagger) with JWT and schema validation.
- Developed backend services with Python, FastAPI, and Django; designed relational schemas, integrated third-party APIs, and collaborated via Git in an Agile team.

ERP Developer (Java) Jun 2023 – Jun 2024
Ramachandran Pvt. Ltd. *Trivandrum, India*

- Built an end-to-end ERP system in Java covering payroll, HR, departments, stock, inventory, and audit modules.
- Developed modules for salary processing, employee records, and stock and inventory control with audit tracking.
- Built a self-billing system to automate bill generation and streamline the billing workflow.

Projects

CargoXFlow – Logistics Management Platform | *Python, FastAPI, PostgreSQL, Redis, WebSockets, Docker*

- Managed the full import and export shipment lifecycle, from booking and documentation to customs clearance, tracking, and billing.
- Implemented a Redis-based distributed locking system that locks a shipment file when a user opens it, preventing concurrent edits, with TTL-based auto-release to avoid stale locks.
- Built real-time updates with WebSockets, broadcasting file lock status, shipment changes, and notifications.
- Designed PostgreSQL schemas for shipments, containers, parties, and documents, exposed through secure REST APIs with JWT authentication and role-based access control.

AI Recruitment Automation Platform | *Python, FastAPI, LLMs, PostgreSQL*

- Developed an AI-powered recruitment platform that parses resumes into structured candidate profiles and ranks candidates against job descriptions using LLM-based matching.
- Automated email ingestion to extract resumes from incoming applications, with backend APIs, authentication, and database integration for multi-step hiring workflows.

AI Chatbot Platform (RAG) | *Python, FastAPI, LLMs, Vector Database*

- Built a platform for creating custom chatbots that answer from a user's website content and uploaded files.
- Implemented the full RAG pipeline: web content extraction, document parsing, text chunking, embedding generation, vector similarity search, and context-grounded LLM responses.

Technical Skills

Languages: Python, Java, JavaScript, SQL
Frameworks: FastAPI, Django, Flask, React
APIs & Real-Time: REST APIs, WebSockets, OpenAPI (Swagger), JWT Authentication
Databases & Caching: PostgreSQL, MySQL Server, MongoDB, Redis (Caching, Distributed Locking)
AI / ML: RAG, LLM Integration, Embeddings, Vector Search, Machine Learning, Transfer Learning
Tools: Git, Docker, Linux, Postman
Concepts: OOP, Data Structures, Database Design, Concurrency Control, Real-Time Systems, RBAC, ERP Systems

Education

SRM University 2018 – 2022
Bachelor of Computer Science (Big Data Analytics) | CGPA: 8.5/10 *Chennai*

- **Accident Location Prediction:** ML model using Kerala Police data to predict accident-prone locations.
- **Hyperspectral Image Classification:** Transfer learning for land-cover classification of satellite imagery.